Hilux Fokou Ngoumo

Research interests: software and vulnerability security, vulnerability data and its evaluation, security of machine learning systems, empirical software engineering

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EDUCATION

MSc, Information Security, ITMO University, St. Petersburg, Russia Sep 2025 to May 2027 (expected)
Weighted average 4.63 / 5.0. Coursework in machine learning, cryptography, network security and cybersecurity analysis. Open Doors scholarship. Thesis: *Detection of Steganographic Content in Digital Images Using Machine Learning Methods*, supervised by Prof. Sergey Bezzateev.

Semester exchange, M.Tech Cyber Security, Parul University, Vadodara, India Aug 2026 to present
Research on vulnerability data and evaluation with Dr. Mukesh Patidar.

BCA (Bachelor of Computer Applications), GITAM University, Bengaluru, India Sep 2021 to Apr 2024
CGPA 8.70 / 10, First Class with Distinction; Outstanding Achiever 2024. Study in India fully funded scholarship (Ministry of Education, India). Thesis on NLP for fake-news and hate-speech detection, supervised by Dr. Nagaraja Perumal.

RESEARCH

- **What the Record Remembers: Snapshot Leakage in Predicting Exploited Vulnerabilities from Public CVE Data.** H. Fokou Ngoumo, M. Patidar. Under review, International Journal of Information Security, 2026. On 183,409 CVEs, post-disclosure fields raise average precision on future CVEs from about 0.08 to 0.85; random cross-validation overstates clean-feature results about threefold against forward-in-time testing. Audit checklist and full reproduction package.
- **Lightweight Open-Set Learning for Zero-Day Network Intrusion Detection: A Compute-Constrained Benchmark on NSL-KDD.** H. Fokou Ngoumo. Under review, ICISSP 2027. Extended version adds CICIoT2023: the best method on NSL-KDD (mean AUROC 0.950) falls to 0.418, while Isolation Forest and PCA reconstruction stay robust on both.
- **NLP in the Digital Age: Combating Fake News, Hate Speech, and Ethical Risks for Social Integrity.** H. Fokou Ngoumo. SSRN preprint, 2025. DOI 10.2139/ssrn.5181145. LSTM and BERT classifiers with bias, robustness and privacy as design requirements.
- **MSc thesis (in progress).** Steganalysis with an SRM-style baseline and an SRNet-style CNN in PyTorch on BOSSBase; cross-method protocol training on LSB and LSB matching and testing on PVD and S-UNIWARD.

TECHNICAL SKILLS

Programming Python, C++, Java, JavaScript, SQL, PHP

Machine learning PyTorch, TensorFlow, scikit-learn, transformers (BERT), LSTM, NLP, computer vision (OpenCV)

Evaluation practice forward-in-time and leave-category-out splits, leakage audits, AUROC / average-precision analysis, reproducible pipelines

Systems and security Git, Docker, Linux, AWS, Azure; vulnerability assessment, penetration testing, cryptography, threat modelling

Languages English and French (native), Russian (B1), German (basic)

SELECTED PROJECTS

- **FakeAlert** (2024 to 2025): fake-news and hate-speech detection with LSTM and BERT, reaching up to 98% accuracy, with bias and adversarial-robustness checks. Basis of the SSRN paper. github.com/hiluxfnh/ITMO_FakeAlert_NLP
- **Sealtrace** and **PhishBlock**: a browser tool that seals scam and harassment screenshots into self-verifying evidence capsules, and a phishing-detection tool for non-technical users.
- **ProctoShield** (2023): exam-integrity system combining OpenCV, a Verhoeff checksum and NLP; cut exam fraud by about 40% in a GITAM pilot. 3rd place, National HackFest India 2023.
- **KamerLark** (2022 to present): AI-assisted student housing platform I founded; 500+ users and a Rs. 100,000 grant.

EXPERIENCE

AI and Software Engineering Intern, 3rd Eye Vision Technology Jul 2022 to Sep 2023
Built a logistics-tracking tool in Python / Django (about 25% workflow gain); prototyped computer-vision models (OpenCV, TensorFlow) for gesture recognition.

Android Developer Intern, PHN Technology Mar 2023 to Jun 2023
Delivered three production Android apps end to end; cut user-reported bugs by about 30% through structured testing.

Youth Panellist and Project Reviewer, Global Youth Mobilization Mar 2024 to present

Reviewed 120+ youth-led project proposals across Africa and Europe against fund criteria, contributing to the allocation of over EUR 1,000,000 in micro-grants.

INTERNATIONAL COLLABORATION AND LEADERSHIP

Young Person Co-Chair and Trainer of Trainers, Inter-Agency Working Group on ASRHR 2022 to 2024
One of four global co-chairs. Trained 36+ youth-led organisations reaching 13,000+ adolescents in humanitarian settings; co-developed a supplementary guide to the 2020 ASRHR Toolkit, published in four languages.

Research and Communication, Women for a Change, Cameroon 2022 to 2026
Research and policy briefs on gender equality and health; amplified campaigns to 500,000+ young people online and co-drove #EndPadTaxNow, an 18-month menstrual dignity campaign.

Team lead, CityMIL, UNESCO Youth Hackathon 2023
Led a five-person team across Cameroon, India, Togo, Bangladesh and Tanzania building a media and information literacy platform. Global Team Award.

International forums
Youth delegate or participant at the UN Commission on the Status of Women (CSW66 to CSW68), COP27, COP28 and SB58; member of the Plant-for-the-Planet Global Ambassador Council (2025 to 2027).

CERTIFICATIONS AND HONOURS

- Certified Ethical Hacker (CEH); ISC2 Certified in Cybersecurity; Google Cybersecurity Professional Certificate; Microsoft SC-900, AZ-900, AI-900; IBM Introduction to AI.
- Open Doors Scholarship, ITMO (2025); UNDP Tech4Peace Cybersecurity Fellow (2025); Plant-for-the-Planet Global Ambassador Council (2025 to 2027); UNESCO Youth Hackathon Global Team Award (2023); Outstanding Achiever, GITAM (2024).

REFERENCES

Dr. Mukesh Patidar
Faculty of Engineering and Technology, Parul University, Vadodara, India. Research supervisor during my exchange and co-author of the CVE snapshot-leakage paper (2026).
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Dr. Pir Mohammad Khan
Faculty of Engineering and Technology, Parul University, Vadodara, India. Supervisor during my exchange at Parul University (2026).
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